HENRIQUE K. MIYAMOTO

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CentraleSupélec, Université Paris-Salcay
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EDUCATION

Université Paris-Saclay, France — 2022–2026
Ph.D. in Information Theory
Thesis: Universal Decoding through Randomised Decoding

Universidade Estadual de Campinas, Brazil — 2021–2022
M.Sc. in Applied Mathematics
Dissertation: Geometry, Statistics and Applications to Communications and Learning

CentraleSupélec, France — 2018–2021
Diplôme d'Ingénieur (M.Sc.)
Major: Applied Mathematics

Universidade Estadual de Campinas, Brazil — 2015–2021
B.Sc. in Electrical Engineering
Class rank: 1/74, GPA: 0.9196/1.0

RESEARCH INTERESTS

Information theory and digital communications (MSC2020 94A and 94B).

RESEARCH EXPERIENCE

2022–present, *doctoral researcher*, **L2S, CNRS/CentraleSupélec/Université Paris-Saclay**, France.

- Advisor: Prof. Sheng Yang.

2021–2022, *master's researcher*, **IMECC, Universidade Estadual de Campinas**, Brazil.

- Advisor: Prof. Sueli I. R. Costa.

2020–2020, *researcher intern*, **Huawei Technologies**, France.

- Supervisors: Dr. Ingmar Land, Dr. Apostolos Destounis and Prof. Jean-Claude Belfiore.

2019–2020, *student researcher*, **L2S, CNRS/CentraleSupélec/Université Paris-Saclay**, France.

- Supervisor: Prof. Sheng Yang.

2016–2018, *undergraduate student researcher*, **IMECC, Universidade Estadual de Campinas**, Brazil.

- Supervisors: Prof. Henrique N. Sá Earp and Prof. Sueli I. R. Costa.

TEACHING TRAINING

Université Paris-Saclay, France — 2022–2026
Certificate "Teaching in Higher Education" (*Parcours "Enseigner dans le Supérieur"*)

- This program certifies training in different teaching skills, and teaching experience of at least 96h, done in parallel to the preparation of the Ph.D. degree.

TEACHING EXPERIENCE

2022–2025, *graduate teaching assistant*, **CentraleSupélec**, France.

- Information theory
- Signal processing
- Communication theory
- Digital signal processing
- Wireless communications

2022–2022, *graduate teaching assistant*, **Universidade Estadual de Campinas**, Brazil.

- Principles of communications I

2021–2021, *undergraduate teaching assistant*, **Universidade Estadual de Campinas**, Brazil.

- Principles of communications I

PREPRINTS

[Z3] H. K. Miyamoto, R. Combes and S. Yang, "Mismatched exponents for deterministic and randomised noise-guessing decoding", *2606.26954*, 2026 . doi: 10.48550/arXiv.2606.26954.

[Z2] H. K. Miyamoto and S. Yang, "Error exponents for randomised list decoding", *arXiv:2601.09519*, 2026 . doi: 10.48550/arXiv.2601.09519.

[Z1] H. K. Miyamoto and S. Yang, "Universal decoding over finite-state additive channels via noise guessing", *arXiv:2501.12971*, 2025 . doi: 10.48550/arXiv.2501.12971.

JOURNAL PAPERS

[J4] H. K. Miyamoto, F. C. C. Meneghetti, J. Pinele and S. I. R. Costa, "On closed-form expressions for the Fisher–Rao distance", *Information Geometry*, vol. 7, pp. 311–354, Nov. 2024, doi: 10.1007/s41884-024-00143-2.

[J3] H. K. Miyamoto, F. C. C. Meneghetti and S. I. R. Costa, "The Fisher–Rao loss for learning under label noise", *Information Geometry*, vol. 6, pp. 107–126, Jun. 2023, doi: 10.1007/s41884-022-00076-8.

[J2] H. K. Miyamoto and S. Yang, "Context-tree-based lossy compression and its application to CSI representation", *IEEE Transactions on Communications*, vol. 70, no. 7, pp. 4417–4428, Jul. 2022, doi: 10.1109/TCOMM.2022.3173002.

[J1] H. K. Miyamoto, S. I. R. Costa and H. N. Sá Earp, "Constructive spherical codes by Hopf foliations", *IEEE Transactions on Information Theory*, vol. 67, no. 12, pp. 7925–7939, Dec. 2021, doi: 10.1109/TIT.2021.3114094.

CONFERENCE PAPERS

[C9] R. Zhang, H. Miyamoto, R. Combes and S. Yang, "An efficient synchronization scheme for distributed bandits", *IEEE International Symposium on Information Theory (ISIT)*, Guangzhou, 2026.

[C8] H. K. Miyamoto and S. Yang, "On α-tilted noise guessing decoding", *IEEE International Symposium on Information Theory (ISIT)*, Guangzhou, 2026.

[C7] H. K. Miyamoto and S. Yang, "Error exponents for randomised list decoding", *IEEE International Symposium on Information Theory (ISIT)*, Guangzhou, 2026.

[C6] H. K. Miyamoto and S. Yang, "On universal decoding over discrete additive channels by noise guessing", *IEEE Information Theory Workshop (ITW)*, pp. 1–6, Sydney, 2025, doi: 10.1109/ITW62417.2025.11240417.

[C5] H. K. Miyamoto and S. Yang "On universal decoding over memoryless channels with the Krichevsky–Trofimov estimator", *IEEE International Symposium on Information Theory (ISIT)*, pp. 1498-1503, Athens, 2024, doi: 10.1109/ISIT57864.2024.10619414.

[C4] F. C. C. Meneghetti, H. K. Miyamoto, S. I. R. Costa and M. H. M. Costa, "Revisiting lattice tiling decomposition and dithered quantisation", *Geometric Science of Information (GSI)*, Saint-Malo, 2023. In: F. Nielsen and F. Barbaresco (eds.), *GSI 2023*, LNCS 14071, pp. 318–327, Springer, 2023, doi: 10.1007/978-3-031-38271-0_319.

[C3] F. C. C. Meneghetti, H. K. Miyamoto and S. I. R. Costa, "Information properties of a random variable decomposition through lattices", *41st International Conference on Bayesian and Maximum Entropy Methods in Science and Engineering (MaxEnt)*, Paris, 2022. In: *Physical Sciences Forum*, vol. 5, no. 1, 2022, doi: 10.3390/psf2022005019.

[C2] H. K. Miyamoto and S. Yang, "A CSI compression scheme using context trees", *International Zurich Seminar on Information and Communication (IZS)*, pp. 24–28, Zurich, 2022, doi: 10.3929/ethz-b-000535273.

[C1] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, "Constructive spherical codes in 2^k dimensions", *IEEE International Symposium on Information Theory (ISIT)*, pp. 1612–1616, Paris, 2019, doi: 10.1109/ISIT.2019.8849464.

DOMESTIC CONFERENCE PAPERS

[D3] H. K. Miyamoto, "Geometria, estatística e aplicações a comunicações e aprendizado"[1], *Proceeding Series of the Brazilian Society of Computational and Applied Mathematics*, vol. 10, no. 1, 2023, doi: 10.5540/03.2023.010.01.0004 (in Portuguese).

[D2] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, "Construção de códigos esféricos usando a fibração de Hopf", *Jornada Nacional de Iniciação Científica (JNIC), 70ª Reunião Anual da SBPC*, Maceió, 2018, ISSN: 2176-1221 (in Portuguese).

[D1] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, "Construction of spherical codes using the Hopf fibration", *XXV Congresso de Iniciação Científica da Unicamp*, Campinas, 2017, doi: 10.19146/pibic-2017-78809.

PATENT APPLICATIONS

[P1] I. Land, H. K. Miyamoto, A. Destounis and J.-C. Belfiore, "Transfer Learning between Neural Networks", WO2022207097 (2022), CN117121020 (2023), EP4309079 (2024).

THESES

[T1] H. K. Miyamoto, "Geometria, Estatística e Aplicações a Comunicações e Aprendizado" (Geometry, Statistics and Applications to Communications and Learning), *M.Sc. dissertation*, Universidade Estadual de Campinas, Campinas, 2022, hdl: 20.500.12733/6633.

SELECTED PRESENTATIONS (WITHOUT PROCEEDINGS)

2nd France PhD Workshop on Information Theory, *oral presentation*, Gif-sur-Yvette, France, 2026.

IEEE European School on Information Theory (ESIT), *poster presentation*, Eindhoven, The Netherlands, 2024.

[1] This is an extended abstract of [T1], published on the occasion of the *Clóvis Caesar Gonzaga Award* (see Awards).

1st France PhD Workshop on Information Theory, *poster presentation*, Palaiseau, France, 2024.

IEEE European School on Information Theory (ESIT), *poster presentation*, Bristol, UK, 2023.

International Conference on Information Geometry for Data Science (IG4DS), *oral presentation*, Hamburg, Germany (virtual), 2022.

International Congress of Mathematicians (ICM), *oral presentation*, Rio de Janeiro, Brazil, 2018.

Latin American Week on Coding and Information (LAWCI), *poster presentation*, Campinas, Brazil, 2018.

Jornada de Matemática, Matemática Aplicada e Educação Matemática (J3M), *oral presentation*, Curitiba, Brazil, 2017.

AWARDS

1st place in **Clóvis Caesar Gonzaga Award** for master's dissertations in applied and computational mathematics, SBMAC, 2023.

Merit Award in Jornada Nacional de Iniciação Científica (JNIC), SBPC, 2018.

2nd place in **Beatriz Neves Award** for undergraduate research in applied and computational mathematics, SBMAC, 2018.

Academic Excellence in Undergraduate Work in Jornada de Matemática, Matemática Aplicada e Educação Matemática (J3M), category Geometry and Topology, PET Matemática/UFPR, 2017.

Scientific Merit in XXV Congresso de Iniciação Científica da Unicamp, PRP/Unicamp, 2017.

Merit Award for the composition produced in Unicamp entrance exam , COMVEST/Unicamp, 2015.

SCHOLARSHIPS

2021–2022, *Master's Degree Scholarship*, São Paulo Research Foundation (FAPESP), 2021/04516-8.

2018–2020, *Eiffel Excellence Scholarship*, French Ministry of Europe and Foreign Affairs.

2016–2018, *Scientific Initiation Scholarship*, São Paulo Research Foundation (FAPESP), 2016/05126-0.

2013–2013, *Junior Scientific Initiation Scholarship*, Brazilian National Council for Scientific and Technological Development (CNPq).

ACADEMIC SERVICE

Scientific societies membership

2025–present, *member*, Société Mathématique de France (SMF).

2025–present, *member*, Société de Mathématiques Appliquées et Industrielles (SMAI).

2020–present, *student member*, Sociedade Brasileira para o Progresso da Ciência (SBPC).

2018–present, *graduate student member*, Institute of Electrical and Electronics Engineers (IEEE).

- **Information Theory Society**, Communications Society.

Reviewing

Journals

- Information Geometry, Springer
- Journal of Communication and Information Systems (JCIS)

Conferences

- IEEE International Symposium on Information Theory (ISIT) [2023–2026]
- Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT) [2023]

Committees and representative positions

2023–2025, *doctoral and postdoc researchers representative*, Laboratory of Signals and Systems (L2S).

2023, *organising committee*, Encontro de Códigos, Reticulados e Informação, IMECC, Unicamp.

2018–2019, *student representative*, Departamento de Comunicações, FEEC, Unicamp.

LANGUAGES

Portuguese	native
English	advanced (Cambridge CAE)
French	advanced
Spanish	intermediate
Japanese	elementary (JLPT N4)

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